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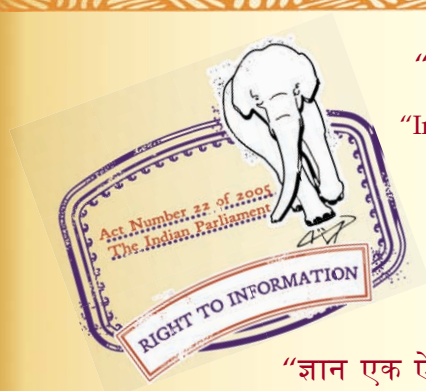
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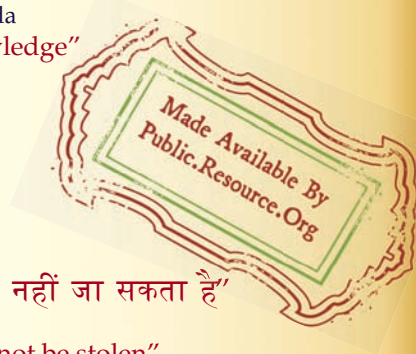
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“Knowledge is such a treasure which cannot be stolen”

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(Reaffirmed 2003)

Indian Standard

**METHODS OF
TEST FOR VULCANIZED RUBBER**

PART 22 CHEMICAL ANALYSIS

(Third Reprint MARCH 1998)

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

Indian Standard

METHODS OF TEST FOR VULCANIZED RUBBER

PART 22 CHEMICAL ANALYSIS

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(Continued on page 2)

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IS : 3400 (Part 22) - 1984

(Continued from page 1)

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(Continued on page 34)

Indian Standard
**METHODS OF
TEST FOR VULCANIZED RUBBER
PART 22 CHEMICAL ANALYSIS**

0. FOREWORD

0.1 This Indian Standard (Part 22) was adopted by the Indian Standards Institution on 13 February 1984, after the draft finalized by the Rubber Product Sectional Committee had been approved by the Petroleum, Coal and Related Products Division Council.

0.2 It is often necessary to identify the rubbers and also to quantify them in vulcanized rubbers. While the identification of rubbers can be done by simple tests, for quantitative analysis detailed chemical investigations are necessary.

0.3 In the preparation of this standard, considerable assistance has been drawn from ASTM D 297-1981 'Rubber products — Chemical analysis' published by the American Society for Testing and Materials, USA.

0.4 A new method namely, 'Amperometric titration method' has been developed in our country for the estimation of free sulphur in unvulcanized compounded rubbers and uncombined sulphur in vulcanizates which is extractable by acetone. This method has been submitted to ISO/TC 45/WG 1 Chemical tests for its consideration.

0.5 For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS : 2-1960*. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1. SCOPE

1.1 This standard (Part 22) prescribes the methods of test for qualitative and quantitative analysis of vulcanized rubbers.

*Rules for rounding off numerical values (revised).

2. QUALITY OF REAGENTS

2.1 Unless otherwise specified, pure chemicals and distilled water (see IS : 1070-1970*) freshly boiled and cooled, shall be employed in tests.

NOTE — ' Pure chemicals ' shall mean chemicals that do not contain impurities which affect the results of analysis.

3. QUALITATIVE ANALYSIS

3.1 Pyrolysis Test

3.1.0 Outline of the Method — Pyrolysis followed by absorption of vapours by specific reagents imparts specific colours to the reagents which help in identification of the sample.

3.1.1 Reagents

3.1.1.1 Solution I — Dissolve 1.0 g of *p*-dimethylamine benzaldehyde and 10 mg of hydroquinone in 100 cm³ of absolute methanol. Add 10 cm³ of ethylene glycol and 5 cm³ of concentrated hydrochloric acid. Dilute the reagent with methanol to obtain relative density of 0.85 at 25°C. The reagent is stable over a period of several months when stored in a brown bottle.

3.1.1.2 Solution II — Dissolve 2.00 g of sodium citrate crystals (2 Na₃C₆H₅O₇ · 11H₂O), 200 mg of citric acid and 30 mg each of indicators bromocresol green and metanil yellow in 500 cm³ of water.

3.1.2 Procedure — Place 0.500 g of sample in a hard glass tube (100 × 75 mm) equipped with a bent glass condensing tube of about 4 mm bore by means of cork stopper. Heat the test tube on a micro-burner or a small flame of Bunsen burner. When the sample starts decomposing immerse the other end of condensing tube into a test tube containing Solution II. If a red colour develops then it confirms polychloroprene rubber. If no colour change is observed then pass the vapours into Solution I and note the colour change. Then add 5 cm³ of methanol and heat on a water bath and note if there is any colour change. Classify the polymer as shown in Table 1.

3.2 Spot Tests

3.2.0 Outline of the Method — Test papers, soaked with specific test solutions are held over the rubber sample being heated. The cracked and volatile fragments of rubber samples escape and imparts characteristic colours to the spot paper.

*Specification for water for general laboratory use (second revision).

TABLE 1 RESULT OF PYROLYSIS TEST
(Clause 3.1.2)

SL No.	TYPE OF POLYMER IN VULCANIZATE	SOLUTION I		SOLUTION II COLOUR
		Initial Colour	Colour After Heating with 5 ml Methanol	
(1)	(2)	(3)	(4)	(5)
i)	Blank	Pale yellow	Pale yellow	Green
ii)	Natural rubber (isoprene rubber)	Brown	Violet blue	Green
iii)	Styrene-butadiene rubber (SBR)	Yellow green	Green	Green
iv)	Nitrile-butadiene rubber (NBR)	Orange red	Red	Green
v)	Chloroprene rubber* (CR)	Yellow	Pale yellow-green	Red
vi)	Butyl rubber (Iso-butylene isoprene rubber) (IIR)	Yellow (droplets float on reagent)	Pale blue-green	Green

NOTE 1 — It is preferable to perform the tests on acetone extracted vulcanizate.

NOTE 2 — In case of blends major portion of the blend would only give these tests.

*Other chlorine containing polymers may give same colour.

3.2.1 Reagents

3.2.1.1 Solution III — Dissolve 2.5 g of benzedene dihydrochloride in a mixture of 500 cm³ methanol and 500 cm³ water. Add 10 cm³ of aqueous dihydroquinone solution (0.1 percent). Store in a brown bottle.

3.2.1.2 Test paper for Solution III — Dissolve 2.0 g of cupric acetate and 0.25 g of metanil yellow in 500 cm³ methanol. Soak filter papers in this solution, dry and cut into strips.

3.2.1.3 Solution IV — Add 5.0 g of yellow mercuric oxide to a mixture of 80 cm³ water and 15 cm³ concentrated sulphuric acid. Heat to boiling and continue heating till the solid completely dissolves. Cool to room temperature and dilute to 100 cm³.

3.2.1.4 Test paper for Solution IV — Blank filter paper strips.

3.2.1.5 Solution V — Dissolve 30 g of trichloroacetic acid with minimum of isopropanol and subsequently dilute to 100 cm³ with the same solvent. (Caution — avoid contact with skin).

3.2.1.6 Test paper for Solution V — Dissolve 3 g of *p*-dimethyl amino-benzaldehyde and 0.05 g of hydroquinone in 100 cm³ of diethyl ether. Soak filter papers in this solution, dry and cut into strips.

3.2.2 Procedure — Wet a strip of each type of test paper with corresponding wetting solution and hold it in parallel position about 5 mm above a heating element that is pressed against the sample. Note the colour change and classify the sample by means of Table 2.

TABLE 2 CLASSIFICATION OF SAMPLES BY SPOT TEST METHOD

SL No.	RUBBER POLYMER	COLOUR OBSERVED IN SPOT TEST PAPERS		
		Solution III	Solution IV	Solution V
(1)	(2)	(3)	(4)	(5)
i)	Chloroprene rubber (CR)	Red	Blank*	Green
ii)	Nitrile-butadiene rubber (NBR)	Green	Brown	Yellow Green
iii)	Styrene-butadiene rubber (SBR)	Blank*	Brown	Blue Green
iv)	Natural rubber (isoprene rubber)	Blank*	Brown	Blue
v)	Butyl rubber (isobutylene isoprene rubber) (IIR)	Blank*	Yellow	Pale Lavender

*Blank colour tests may be pale brown or colourless.

4. QUANTITATIVE ANALYSIS

4.1 Rubber Polymer Content (Indirect Method)

4.1.0 The indirect method determines the non-rubber constituents of a product and the remainder is expressed as rubber polymer content. The non-rubber constituents are determined in accordance with Appendix A (A-1 through A-10 and rubber polymer content is calculated as given in A-12.1). The method is applicable to NR, IR, SBR and BR products and also to IIR products if they are extracted with methyl ethyl ketone rather than with acetone.

NOTE — The abbreviations NR, IR, IIR, SBR, BR, CR and NBR used above and hereafter denote isoprene (natural), isoprene (synthetic), styrene butadiene isobutylene-isoprene, butadiene, chloroprene and nitrile-butadiene rubbers.

4.2 Silicon Dioxide and Insoluble Matter

4.2.1 Reagents

4.2.1.1 Hydrochloric acid

4.2.1.2 Nitric acid

4.2.1.3 Sulphuric acid

4.2.1.4 Hydrogen fluoride — aqueous solution.

4.2.2 Procedure — Ignite a specimen of the sample as indicated in **A-8**. Dissolve the ash in 10 cm³ of hydrochloric acid, rinse the crucible thoroughly, dilute to 100 cm³ and evaporate to dryness in a casserole. Bake for 1 h at 110°C. Moisten with 10 cm³ of hydrochloric acid and 3 drops of nitric acid and digest on steam bath for 15 minutes. Add 100 cm³ of water, boil, filter and wash with hot water. Dry and ignite the precipitate in a porcelain crucible. Save the filtrate for lead oxide determination. Weigh to determine silicon dioxide and insoluble matter. If the residue is large enough to justify an analysis of silicon dioxide. Transfer to a platinum crucible and add 2 to 3 cm³ of hydrogen fluoride and a few drops of sulphuric acid. Evaporate to dryness, and carefully ignite at low red heat. Weight till constant mass is obtained. The loss in mass is silicon dioxide.

4.2.3 Calculation

$$\text{Silicon dioxide and insoluble matter, percent by mass} = \frac{(A - B) \times 100}{C}$$

$$\text{Silicon dioxide, percent by mass} = \frac{(A - B) - (D - E)}{C} \times 100$$

where

A = mass in g of residue and porcelain crucible after ignition,

B = mass in g of porcelain crucible,

C = mass in g of original specimen,

D = mass in g of residue and platinum crucible after treatment with hydrogen fluoride and ignition, and

E = mass in g of platinum crucible.

4.3 Lead Oxide

4.3.1 Reagents

4.3.1.1 Ethanol — 95 percent (v/v).

4.3.1.2 Ethanol — 50 percent (*v/v*).

4.3.1.3 Hydrogen sulphide — gas and saturated solution.

4.3.2 Procedure — Just neutralise the filtrate obtained in 4.2.2 with ammonium hydroxide and add 1 cm³ of hydrochloric acid. Pass hydrogen sulphide into the solution until precipitation is complete and dilute to 50 to 100 cm³. Filter and wash with a saturated solution of hydrogen sulphide. Save the filtrate for determination of iron and aluminium oxide. Dissolve lead sulphide in nitric acid (1 : 1), boil to complete solution. Filter and cool the filtrate. Add 10 cm³ of sulphuric acid and evaporate to dense white fumes of sulphuric acid. Cool, dilute with 50 cm³ of water, add an equal volume of ethanol (95 percent) and let stand overnight. Filter on a tared Gooch crucible, wash with ethanol (50 percent) and dry at 105°C.

4.3.3 Calculation

$$\text{Lead oxide, percent by mass} = \frac{(A - B) \times 0.736 \times 100}{C}$$

where

A = mass in g of crucible and lead sulphate,

B = mass in g of crucible,

C = mass in g of original specimen, and

0.736 = conversion factor from lead sulphate to lead oxide.

4.4 Iron and Aluminium Oxides (R_2O_3)

4.4.1 Reagents

4.4.1.1 Ammonium chloride — solid.

4.4.1.2 Ammonium chloride solution — 20 g/cm³.

4.4.1.3 Methyl red indicator solution

4.4.1.4 Potassium ferricyanide indicator

4.4.2 Procedure — Boil the filtrate obtained in 4.3.2 from the lead sulphide precipitation to expel hydrogen sulphide. Adjust the volume of solution to 100 to 150 cm³. Add a few drops of nitric acid and boil the solution again. Test for ferrous iron, using potassium ferricyanide as an outside indicator on a spot plate. If ferrous iron is present, add more nitric acid and proceed as before until all the iron is oxidised. Add 5 g of solid ammonium chloride. Add ammonium hydroxide until the solution is coloured definitely yellow by methyl red, but do not add an excess. Heat to boiling and boil for 5 minutes. When the

precipitate has settled, filter, and wash with ammonium chloride solution. Save the filtrate for determination of calcium oxide. Carefully char off the filter paper at low temperature and ignite in free oxidizing atmosphere.

4.4.3 Calculation

$$\begin{array}{l} \text{Iron and aluminium oxides,} \\ \text{percent by mass} \end{array} = \frac{(A - B)}{C} \times 100$$

where

A = mass in g of crucible including ash,

B = mass in g of crucible, and

C = mass in g of original sample.

4.5 Calcium Oxide (Volumetric Method)

4.5.1 Reagents

4.5.1.1 *Ammonium oxalate* — saturated solution.

4.5.1.2 *Ammonium sulphide*

4.5.1.3 *Formic acid mixture* — 200 cm³ of formic acid (relative density 1.22), 770 cm³ of water and 30 cm³ of ammonium hydroxide.

NOTE — Unopened bottles of formic acid may have sufficient pressure to cause an explosion as the top is removed protective clothing and guard should be used when a bottle is first opened. Do not allow formic acid to come in contact with the skin.

4.5.1.4 *Formic acid wash solution* — Dilute 30 cm³ of formic acid mixture to 1 dm³ and saturate with hydrogen sulphide.

4.5.1.5 *Hydrogen sulphide*

4.5.1.6 *Methyl red indicator solution*

4.5.1.7 *Oxalic acid solution* — 100 g/dm³.

4.5.1.8 *Oxalate wash solution* — (2 g ammonium oxalate + 1 g oxalic acid) mixture per dm³.

4.5.1.9 *Potassium permanganate* — 0.1 N.

4.5.2 Procedure

4.5.2.1 Dilute the filtrate obtained in 4.4.2 to 200 cm³, neutralise with hydrochloric acid (1 : 1) and add 10 cm³ in excess. Heat to boiling. Add sulphuric (1 : 36) to precipitate barium, but avoid a large excess. Digest until the precipitate settles, test for completeness of precipitation and digest for at least 2 hours or until the precipitate is filterable. Filter,

IS : 3400 (Part 22) - 1984

wash and discard the precipitate. Concentrate the filtrate to 200 cm³, neutralise with ammonium hydroxide to methyl red indicator neutralise with sulphuric acid (1 : 37) and add 25 cm³ of formic acid mixture.

4.5.2.2 Heat the solution to 60°C and saturate with hydrogen sulphide for 20 minutes. Digest for 1 hour at 60°C, filter, wash with formic acid wash solution. Test for complete removal of zinc with ammonium sulphide in alkaline solution and filter, if necessary. Make the filtrate just acidic with hydrochloric acid and evaporate to 150 cm³. Filter to remove sulphur. Neutralize with ammonium hydroxide and add 1 ml excess. Make solution just acidic with oxalic acid solution and add 12 cm³ in excess and boil for 2 minutes while stirring. Add 50 cm³ of saturated ammonium oxalate solution, add more if solution is still acidic to methyl red. Dilute to 250 cm³, boil for 2 minutes and digest on steam bath for 1 hour. Cool, filter and wash with oxalate wash solution. Save filtrate uniting with next filtrate and washings (after reprecipitation) for determination of magnesium oxide. Dissolve the precipitate in 50 cm³ of warm hydrochloric acid (1 : 3) and reprecipitate by the above procedure.

4.5.2.3 Wash the precipitate with water dissolve the same in hot sulphuric acid (1 : 5) and titrate hot with standard potassium permanganate.

4.5.3 Calculation

$$\text{Calcium oxide, percent by mass} = \frac{(A \times N \times 0.028)}{C} \times 100$$

where

A = volume in cm³ of potassium permanganate solution required for titration,

N = normality of potassium permanganate solution,

C = mass in g of original specimen, and

0.028 = grams of calcium oxide equivalent to each cm³ of 1 N potassium permanganate solution.

4.6 Calcium Oxide (Gravimetric Method)

4.6.1 Alternatively calcium oxide can be determined gravimetrically by drying and igniting the precipitate obtained in 4.5.2.2 in a heat resistant covered porcelain crucible — the ignition temperature being 1 000 to 1 200°C.

4.6.2 Calculation

$$\text{Calcium oxide, percent by mass} = \frac{A - D}{C} \times 100$$

where

A = mass in g of the precipitate plus crucible,

D = mass in g of the crucible, and

C = mass in g of original specimen.

4.7 Magnesium Oxide**4.7.1 Reagent**

4.7.1.1 Ammonium phosphate solution — 100 g/dm³.

4.7.2 Procedure

4.7.2.1 Evaporate to dryness the filtrate and washings obtained in 4.5.2.2. Add 50 cm³ of nitric acid, cover, and warm till evolution of gas subsides. Uncover, evaporate to dryness. Heat the residue on hot plate for 2 to 3 hours, overnight if convenient. Dissolve in 100 cm³ of water, slightly acidify with hydrochloric acid and add 25 cm³ of ammonium phosphate solution. Cool to 15°C (use ice bath), neutralise slowly with ammonium hydroxide while stirring using methyl red as an indicator and add 10 cm³ of ammonium hydroxide in excess. Let it stand for overnight, filter (without attempting to transfer precipitate) and wash with ammonium hydroxide.

4.7.2.2 Dissolve the precipitate in warm hydrochloric acid using the same beaker of precipitate. Dilute to 100 cm³, add 5 cm³ of ammonium phosphate solution, neutralise slowly with ammonium hydroxide while stirring and add 5 cm³ in excess. Let it stand for at least 4 hours. Filter through a filter paper preferably charred off at low temperature and ignite at 1 000 to 1 200°C for 60 minutes and weigh.

4.7.3 Calculation

$$\text{Magnesium oxide, percent by mass} = \frac{(A - B) \times 0.3621}{C} \times 100$$

where

A = mass in g of crucible and precipitate,

B = mass in g of crucible,

C = mass in g of original specimen, and

0.3621 = conversion factor for magnesium pyrophosphate to magnesium oxide.

4.8 Zinc Oxide

4.8.1 Apparatus

4.8.1.1 *Muffle furnace* — Capable of being heated to $550 \pm 25^{\circ}\text{C}$.

4.8.1.2 *Burette* — 10 cm^3 , graduated in 0.02 cm^3 divisions.

4.8.1.3 *Burette* — 50 cm^3 , graduated in 0.1 cm^3 divisions.

4.8.2 Reagents

4.8.2.1 *Acetone*

4.8.2.2 *Aluminium chloride solution* — 0.1 M .

4.8.2.3 *Ammonium fluoride solution* — 3 M (store in a polyethylene or wax coated bottle).

4.8.2.4 *Buffer solution* — 60 g of acetic acid and 77 g of ammonium acetate dissolved in water and diluted to 1 dm^3 .

4.8.2.5 *Dithiozone indicator solution* — 0.01 g dithiozone in 10 cm^3 of acetone.

4.8.2.6 *EDTA solution* — 0.01 M .

4.8.2.7 *Methyl orange solution* — 1 g/dm^3 in water.

4.8.2.8 *2, 4-Pentandione solution* — 10 cm^3 in 90 cm^3 acetone.

4.8.2.9 *Zinc chloride solution*

4.8.2.10 *Magnesium chloride solution* — 0.1 M .

4.8.3 Procedure

4.8.3.1 Weigh 1 g of sample in 50 cm^3 porcelain crucible and ash it. Cool the crucible, wash the ash in a beaker with water. Add 5 cm^3 of hydrochloric acid to the crucible and warm it. Pour the washings into the beaker. Rinse the crucible with 5 cm^3 of hydrochloric acid. Wash. Do not filter the solution. Add 10 cm^3 of hydrochloric acid to the beaker. Evaporate the solution to 10 cm^3 and make up the solution to 100 cm^3 . Agitate the solution until boiling begins by covering the top of the hot plate with sand.

4.8.3.2 Select an aliquot as indicated below and transfer it to a 250 cm³ conical flask, mixing the solution and suspended solids well before aliquoting.

<i>Zinc Oxide Expected</i>	<i>Aliquot Size</i>	<i>Burette Size</i>
percent	cm ³	cm ³
0 to 3	25	10
3 to 8	10	10
Above 8	10	50

4.8.3.3 Dilute the aliquot to 25 cm³, add 2 cm³ of aluminium chloride solution 10 cm³ of aluminium fluoride solution and a drop of methyl orange indicator solution. Add ammonium hydroxide till colour changes to orange yellow, add 1 cm³ of hydrochloric acid and bring the solution to a boil. Cool. Add ammonium hydroxide till colour becomes yellow. Add 0.5 cm³ more ammonium hydroxide. Add 10 cm³ of buffer solution, 60 cm³ of acetone and 5 drops of dithiozone indicator solution. Add 5 cm³ of 2, 4-pentandione solution. Cool in an icebath before titration. Titrate with EDTA solution to a yellow-green colour using the burette selected as in **4.8.3.2**.

4.8.4 Calculation

$$\text{Zinc oxide, percent by mass} = \frac{V \times C \times 100 \times 100}{S \times A}$$

where

V = volume in cm³ of EDTA solution used in titration,

C = concentration of EDTA solution in g of zinc oxide/cm³,

S = mass in g of the sample taken, and

A = aliquot size in cm³.

4.9 Total Barium as Barium Sulphate

4.9.1 Reagents

4.9.1.1 *Hydrogen sulphide*

4.9.1.2 *Sodium carbonate solid*

4.9.1.3 *Sodium carbonate solution* — 50 g/dm³.

4.9.1.4 *Sodium nitrate*

4.9.2 Procedure — Fuse the ash specimen with 5 g of mixture of equal parts of sodium carbonate and sodium nitrate. Stir well during fusion. Cool, place in a 400 cm³ beaker with about 125 cm³ of water and digest on water bath overnight. Filter and wash the residue well with hot sodium carbonate solution. Wash this residue back into same beaker with hot water, dissolve the residue in the beaker and any traces on the filter paper with hydrochloric acid and heat the solution on the steam bath. Filter and wash thoroughly with hot water. Adjust acidity by ammonium hydroxide (2 : 3) and hydrochloric acid (1 : 1) to provide an excess of 4 to 5 cm³ of hydrochloric acid per 100 cm³ of solution. Cool, saturate the solution with hydrogen sulphide to settle lead sulphide and filter into 400 cm³ beaker and wash thoroughly to total volume of 200 cm³ or less. Precipitate barium with sulphuric acid (1 : 1) and treat the precipitate in the usual manner as for sulphur determination.

4.9.3 Calculation

$$\begin{array}{l} \text{Total barium as barium sulphate,} \\ \text{percent by mass} \end{array} = \frac{A}{B} \times 100$$

where

A = mass in g of barium sulphate, and

B = mass in g of specimen taken.

4.10 Total Barium as Barium Carbonate

4.10.1 Reagents

4.10.1.1 Ammonium carbonate

4.10.1.2 Carbon dioxide

4.10.1.3 Hydrogen sulphide

4.10.2 Procedure — Place 1 g of specimen in a porcelain boat, and place this in a combustion tube through which passes a current of carbon dioxide. Ash the sample in the tube. After ignition and cooling in the carbon dioxide atmosphere, remove and grind the residue and transfer in a beaker. Treat with 5 to 10 g of ammonium carbonate, 15 to 20 ml of ammonium hydroxide and about 50 ml of water. Boil for 20 minutes, filter and wash. Transfer precipitate into original beaker and add 10 ml of glacial acetic acid with sufficient water to make volume to about 100 ml. Heat to boil, filter through the same filter paper. Pass hydrogen sulphide into the filtrate to precipitate lead. Filter, wash, and discard the precipitate. Precipitate barium with sulphuric acid (1 : 16) and determine in the usual manner.

4.10.3 Calculation

$$\text{Barium carbonate, percent by mass} = \frac{A \times 0.8458}{B} \times 100$$

where

A = mass in g of barium sulphate, and

B = mass in g of the specimen taken.

4.11 Total Antimony**4.11.1 Reagents**

4.11.1.1 *Hydrogen sulphide*

4.11.1.2 *Methyl red indicator solution*

4.11.1.3 *Potassium bromate standard solution — 0.1 N.*

4.11.1.4 *Potassium sulphate*

4.11.1.5 *Sodium sulphite*

4.11.1.6 *Starch iodate paper*

4.11.2 Procedure

4.11.2.1 Weigh 0.5 g of sample and transfer to a Kjeldahl's flask. Add 25 ml of sulphuric acid, 10 to 12 g of potassium sulphate, place funnel in the neck of flask and heat till colourless solution is obtained. Cool, wash the funnel, dilute to 100 ml with water and transfer to 400 ml beaker. Dilute to 250 ml with hot water and precipitate antimony with hydrogen sulphide. Filter, transfer the precipitate to a Kjeldahl's flask. Add 15 ml of sulphuric acid, 10 to 12 g of potassium sulphate and heat till colourless solution is obtained.

4.11.2.2 Wash the funnel, dilute to 100 ml, add 1 to 2 g of sodium sulphite and boil until all sulphur dioxide is driven out. (No blue colour with starch iodate paper indicator). Add 25 ml of hydrochloric acid, dilute to 200 ml, regulate temperature to 60°C, add 2 drops of 0.2 percent methyl red solution and titrate with 0.1 N potassium bromate solution till the solution is colourless. When indicator starts to fade, add potassium bromate slowly, using another drop of indicator. At the end point an added drop of indicator should become colourless.

IS : 3400 (Part 22) - 1984

4.11.3 Calculation

$$\begin{array}{l} \text{Antimony (as Sb}_2\text{S}_3 \text{), percent} \\ \text{by mass} \end{array} = \frac{(A \times N) \times 0.0849}{B} \times 100$$

where

A = volume in ml of potassium bromate solution required for titration of specimen,

B = mass in g of specimen used, and

N = normality of potassium bromate solution.

4.12 Titanium Dioxide

4.12.1 Reagents

4.12.1.1 *Hydrogen peroxide* — 3 percent.

4.12.1.2 *Potassium pyrosulphate powder*

4.12.1.3 *Titanium dioxide*

4.12.1.4 *Phosphoric acid*

4.12.2 Apparatus

4.12.2.1 *Spectrophotometer or filter photometer* — Suitable for measurement of absorbance at approximately 416 with matched absorption cells 1 to 3 cm in thickness. The instrument shall be suitably calibrated.

4.12.3 Procedure

4.12.3.1 Weigh 50 mg of rubber sample in a platinum crucible. Place in a cold muffle furnace and heat to 550°C. Remove, cool and add 2 g of fused potassium pyrosulphate powder and 2 to 3 drops of sulphuric acid to ashed crucible and a blank crucible. Heat gently until the pyrosulphate is melted. Gradually increase the heat till dull red and continue heating for 10 to 15 minutes. Cool while rolling the crucible.

4.12.3.2 Place crucible in a 50 cm³ beaker and cover with sulphuric acid (6 : 100). Cover with watch glass, boil slowly till all matter except clay or silica have dissolved. Remove heat and remove crucible from beaker, wash with sulphuric acid. Filter, if necessary and transfer quantitatively the solution to 50 cm³ volumetric flask using sulphuric acid for rinsing the beaker and filter. Add 2 cm³ of phosphoric acid and 5 cm³ of hydrogen peroxide to the flask and dilute to volume with sulphuric acid. Mix and allow to stand for 5 minutes. Measure the absorbance of unknown and blank at 416 mm using sulphuric acid in the reference cell of the spectrophotometer

or filter photometer. If the absorbance is greater than 1.5, add 2 cm³ of hydrogen peroxide to an aliquot of the solution and dilute to known volume with sulphuric acid. Measure absorbance at 416 nm. If the absorbance is less than 0.150, a large specimen shall be used.

4.12.4 Calculation

4.12.4.1 For spectrophotometer

$$\text{Titanium dioxide, percent by mass} = \frac{(A_s - A_b)}{aE} \times 100$$

where

A_s = absorbance of specimen solution,

A_b = absorbance of blank solution,

a = absorptivity of titanium dioxide, and

E = milligrams of specimen per ml of solution on which A_s is measured.

4.12.4.2 For filter photometer

$$\text{Titanium dioxide, percent by mass} = \frac{A}{B} \times 100$$

where

A = mg of titanium dioxide per ml of solution, and

B = mg of specimen per ml of same solution.

APPENDIX A

(Clause 4.1)

DETERMINATION OF RUBBER POLYMER CONTENT (INDIRECT METHOD)

A-1. TOTAL EXTRACT

A-1.1 Apparatus

A-1.1.1 *Extraction Apparatus* — See Fig. 1.

A-1.2 Reagents

A-1.2.1 *Acetone* — distilled over potassium carbonate boiling between 56 to 57°C.

A-1.2.2 *Chloroform*

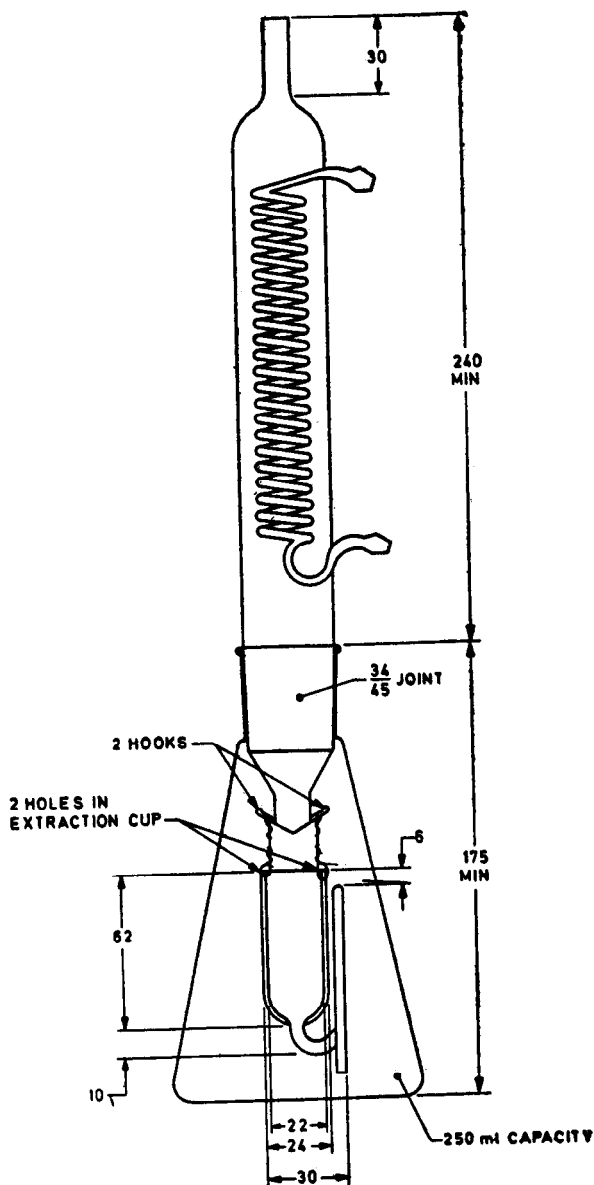


FIG. 1 EXTRACTION APPARATUS WITH GLASS CONDENSER AND GROUND JOINTS

A-1.3 Procedure

A-1.3.1 Place a weighed specimen of approximately 2 g in a filter paper. If the specimen is in the form of a sheet, cut it with scissors into strips 3 to 5 mm in width. If the specimen becomes tacky during the extraction, take care that adjacent portions are separated by paper. Fold the paper so that it will fit in the extraction cup, and suspend the cup in a weighed extraction flask containing 50 to 75 ml of a mixture consisting of 32 parts of acetone and 68 parts of chloroform by volume. (Prior to the weighing of the extraction flask, it shall have been dried for 2 h at $70^{\circ}\text{C} \pm 5^{\circ}\text{C}$ and cooled in a desiccator to the temperature of the balance.)

A-1.3.2 Extract the specimen continuously for 16 h, heating at a rate such that the time required to fill and empty the siphon cup will be between 2.5 and 3.5 min. (Rubber products having a ratio of total sulphur to rubber polymer in excess of 10 percent shall be extracted for 72 h). Carefully note all characteristics of the extract, both when hot and cold. If the colour is black, make a chloroform extraction separately for 4 h, dry the extract to constant mass and add the value to the result obtained for total extract.

A-1.3.3 Evaporate off the solvent over a steam bath, using a gentle current of filtered air to prevent boiling. Remove the flask from the steam bath just prior to the disappearance of the last traces of solvent to prevent loss of extract. Continue the passage of air through the flask for 10 min to remove the remaining solvent and dry the flask for 2 h in a $70^{\circ}\text{C} \pm 5^{\circ}\text{C}$ air bath.

A-1.3.4 Cool in a desiccator to the temperature of the balance and weigh. Save the extracted rubber for further tests that require the use of an extracted specimen.

A-1.4 Calculation

$$\text{Total extract, percent} = \frac{A}{B} \times 100$$

where

A = mass in g of extract, and

B = mass in g of specimen used.

A-2. ALCOHOLIC POTASH EXTRACT

A-2.1 Reagent

A-2.1.1 Alcoholic Potash Solution — Prepare a 1 N alcoholic potassium hydroxide solution by dissolving the required amount of potassium hydroxide in absolute ethanol that has been purified as follows:

Dissolve 1.5 g of silver nitrate in 3 ml of water and add it to one litre of alcohol. Dissolve 3 g of potassium hydroxide in the smallest

amount of hot water, cool, add it to the silver nitrate solution, and shake thoroughly. Allow the solution to stand for at least 24 h, filter, and distil.

NOTE — Absolute ethanol denatured with 10 percent by volume of methanol may also be used.

A-2.1.2 Congo Red Paper

A-2.1.3 Ethanol — 95 percent.

A-2.1.4 Ether

A-2.2 Procedure

A-2.2.1 Remove the specimen remaining after the determination of total extract from its wrapping material while wet with solvent and dry the rubber at $70 \pm 5^\circ\text{C}$ to remove the solvent.

A-2.2.2 Transfer to a 200 ml Erlenmeyer flask, add 50 ml of alcoholic potash solution, and heat under a reflux condenser for 4 h. In the case of hard rubber, continue the heating for 16 h or more.

A-2.2.3 Filter into a 250 ml beaker, wash with two 25 ml portions of boiling alcohol and then with three 25 ml portions of boiling water, and evaporate the filtrate just to dryness. Use about 75 ml of water to transfer the dried filtrate residue to a separatory funnel. Acidify the solution with hydrochloric acid (1 : 3), testing with Congo red paper.

A-2.2.4 Extract with four 25 ml portions of ether, unless the fourth portion should be coloured, when the extraction shall be continued until no further quantity can be removed. Unite the ether fractions and wash thoroughly with water until free of acid (two washings are generally sufficient).

A-2.2.5 Filter the ether solution through a plug of previously washed absorbent cotton into a weighed flask and wash the separatory funnel and the cotton plug with ether. Evaporate the ether on a steam bath, using a gentle current of filtered air to prevent boiling. Remove the flask from the steam bath just prior to the disappearance of the last traces of solvent and continue the passage of air for 10 minutes. Dry the flask at $100^\circ\text{C} \pm 5^\circ\text{C}$ to constant mass, cool and weigh.

A-2.3 Calculation

$$\begin{array}{l} \text{Alcoholic potash extract,} \\ \text{percent by mass} \end{array} = \frac{A}{B} \times 100$$

where

A = mass in g of the extract, and

B = mass in g of specimen used.

A-3. TOTAL SULPHUR (ZINC-NITRIC ACID METHOD)

A-3.0 General — This method covers the determination of all the sulphur, except that contained in barium sulphate, in a sample of rubber product or in the fillers obtained from a rubber product. If acid-soluble barium salts, antimony sulphide, or lead compounds are present, this method will give erroneous results, in which case the fusion method (*see A-4*) should be used. This method may be used for the determination of organic plus inorganic sulphur on an extracted sample, total plus inorganic sulphur on an unextracted sample, or inorganic sulphur in fillers. If it is used for determination of total or organic sulphur plus inorganic sulphur, it shall also be used for determination of inorganic sulphur (*see A-5*). The method is applicable to NR, SBR, BR, IR and OR products and to total plus inorganic sulphur, determination of NBR products.

A-3.1 Reagents

A-3.1.1 Barium Chloride Solution — (100 g $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ /litre) — Dissolve 100 g of barium chloride ($\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$) in water, and two to three drops of hydrochloric acid, and dilute to 1 litre. If there is any insoluble matter or cloudiness, heat the solution overnight on a steam bath and filter.

A-3.1.2 Bromine, Saturated Water Solution

A-3.1.3 Nitric Acid, Fuming

A-3.1.4 Picric Acid, Saturated Solution

A-3.1.5 Potassium Chlorate — (KClO_3)

A-3.1.6 Zinc-Nitric Acid Solution — Add 200 g of zinc oxide (ZnO) to one litre of nitric acid slowly and with caution against spattering.

A-3.2 Procedure

A-3.2.1 Place 0.5 g of soft rubber or 0.2 g of hard rubber (extract the specimen with acetone or acetone-chloroform mixture if organic sulphur is to be determined) in a 500 cm^3 conical destruction flask of chemically resistant material. Add 10 cm^3 of zinc-nitric acid solution and moisten the sample thoroughly. Let stand at least for 1 h; overnight if convenient. By so doing the sample becomes partly decomposed; this permits the addition of fuming nitric acid with no danger of ignition of the sample. Add 15 cm^3 of fuming nitric acid and whirl the flask rapidly to keep the sample immersed to avoid ignition. With some samples it may be necessary to cool the flask under running water.

A-3.2.2 When the dissolution of the rubber appears to be complete, add 5 cm^3 of a saturated water solution of bromine and slowly

evaporate the mixture to a foamy syrup. (For the determination of sulphur in unvulcanized mixtures, use 3 cm³ of bromine in place of bromine water.)

A-3.2.3 If organic matter or carbon remains at this point, add a few cm³ of fuming nitric acid and a few crystals of potassium chlorate (caution) and evaporate at a boil. Repeat this operation until all carbon is removed and the solution is clear, colourless, or light yellow.

A-3.2.4 At this point either of the following procedures may be used:

Procedure A — Place the flask on an asbestos gauze and evaporate the mixture to dryness over a burner. Then bake the mixture at the highest temperature of the burner until all nitrates are decomposed and no more nitrogen oxide fumes can be detected. The flask and its contents shall be carefully annealed after this procedure by gradually decreasing the flame or by placing the flask on successively cooler sources of heat.

Procedure B — Evaporate the mixture, cool, add 10 ml of hydrochloric acid, and evaporate to dryness, avoiding spattering. Repeat this procedure once, or more than once if oxides of nitrogen are still evolved.

A-3.2.5 Cool the flask, add 50 ml of hydrochloric acid (1 : 6) and digest hot until dissolution is as complete as possible. Filter while hot. Wash the filter and dilute the filtrate and washings to about 300 ml. Add 10 ml of saturated picric acid solution, heat to 90°C, and precipitate the sulphate by dropwise addition of 10 cm³ barium chloride solution while stirring vigorously. Digest the precipitate overnight, preferably at 60 to 80°C, using a watch glass to cover the beaker. Filter the barium sulphate and wash with water until the filter is colourless. Dry, ash, and finally ignite the precipitate at 650 to 900°C, with free access to air, to constant mass. Cool in a desiccator and weigh.

A-3.2.6 Calculation

$$\text{Sulphur, percent} = \frac{A \times 0.1373}{B} \times 100$$

where

A = mass in g of barium sulphate, and

B = mass in g of specimen used.

A-4. TOTAL SULPHUR (FUSION METHOD)

A-4.1 General — This method covers the determination of organic, total or inorganic sulphur in rubber compounds when acid-soluble barium salts, antimony sulphide, or lead compounds are present. All of the sulphur in the sample, including that present in barium sulphate is determined when this method is used. When this method is used for determination of organic or total sulphur it shall also be used for the determination of inorganic sulphur (*see* A-5). Total plus inorganic sulphur shall be determined on an unextracted specimen; organic plus inorganic sulphur on an extracted specimen. The method is applicable to NR, SBR, BR, IR products and to the determination of total plus inorganic sulphur on NBR products.

A-4.2 Reagents

A-4.2.1 Nitric Acid-Bromine Solution — Add a considerable excess of bromine to nitric acid so that a layer of bromine is present in the reagent bottle. Shake thoroughly and allow to stand 24 h before using.

A-4.2.2 Sodium Carbonate (Na_2CO_3)

A-4.2.3 Sodium Carbonate Solution — 50 g/litre.

A-4.3 Procedure

A-4.3.1 Place 0.5 g of rubber or 0.2 g of hard rubber in a porcelain crucible of about 75 ml capacity. The specimen shall have been extracted with acetone or acetone-chloroform mixture (*see* A-1) if organic plus inorganic sulphur is to be determined. Add 15 ml of the nitric acid-bromine mixture, cover the crucible with a watch glass, and let it stand for 1 h in the cold. Heat for 1 h on the steam bath, remove the cover, rinse it with a little water, and evaporate to dryness.

A-4.3.2 Add 3 ml of nitric acid, cover, warm for a short time on the steam bath and then allow to cool. Carefully add in small portions, by means of a glass spatula, 5 g of sodium carbonate. Raise the watch glass only high enough to permit the introduction of the spatula. Allow the sodium carbonate to slide down the side of the crucible, as it shall not be dropped directly into the acid. Rinse the watch glass with 2 or 3 ml of hot water and stir the mixture thoroughly with a glass rod. Digest for a few minutes, spread the mixture halfway up the side of the crucible to facilitate drying, and dry on a steam bath. Fuse the mixture by heating over a sulphur-free flame.

A-4.3.3 Place the crucible in an inclined position on a wire triangle and start the ignition over a low flame. The tendency for the organic matter to burn too briskly may be controlled by judicious use of the

stirring rod with which the burning portion is scraped away from the rest. When part of the mass is burned white, work a fresh portion into it until all of the organic matter is destroyed. It is necessary to hold the edge of the crucible with tongs. Towards the last half of the operation the flame should be increased. It is unnecessary to heat the crucible to redness. With care, a crucible can be used for at least 10 to 12 fusions.

A-4.3.4 After a fusion, allow the crucible to cool. Place it in a 400 ml beaker, add sufficient water to cover the crucible (about 125 ml), and digest on the steam bath or plate for at least 2 h.

A-4.3.5 Filter the solution into a covered 400 ml beaker containing 5 ml of hydrochloric acid and wash the residue thoroughly with hot sodium carbonate solution. A qualitative test for barium may be made on the residue, but no analysis for barium or correction because of its presence is necessary, unless a detailed ash analysis is desired. Acidify the filtrate to indicator paper with hydrochloric acid and add 2 ml excess. Precipitate barium sulphate and complete the determination as described in A-3.

A-4.3.6 Calculation

$$\text{Sulphur, percent by mass} = \frac{A \times 0.1373}{B} \times 100$$

where

A = mass in g of barium sulphate, and

B = mass in g of specimen used.

A-5. INORGANIC SULPHUR

A-5.1 General Type — This method covers the determination of inorganic sulphur in rubber products when no antimony is present. The sample is ashed and sulphur determined in the ash by the fusion method (see A-4) if acid-soluble barium salts or lead are present, or by the zinc nitric acid method in the absence of acid-soluble barium salts. The same method shall be used for the inorganic sulphur determination is used for the determination of organic sulphur or total sulphur. The method is applicable to NR, SBR, BR, IR, CR, NBR and IIR products.

A-5.2 Reagents

A-5.2.1 Nitric Acid-Bromine Solution

A-5.3 Procedure — Extract a 1.0 g specimen with acetone or with acetone-chloroform mixture (see A-1). Dry the sample, place in a

porcelain crucible of about 75 ml capacity and distil off the rubber in a muffle furnace, using a maximum temperature of 450°C. A burner may be used for ashing if the sample is not allowed to catch fire. A wire gauze under the crucible will aid in preventing combustion. The carbon need not be completely burned off in this ignition. If acid soluble barium salts and lead are absent, add 3 ml of nitric acid bromine mixture to the ash, cover with a watch glass, and heat for 1 h. Transfer the contents of the crucible, with washing, into a 500 ml conical destruction flask of chemically resistant material and evaporate to dryness. Proceed with the determination of sulphur as described in A-3.2.3 to A-3.2.5. In the presence of acid-soluble barium salts and lead, determine the sulphur by treating the ash in accordance with A-4.

A-5.4 Calculation

$$\begin{array}{l} \text{Inorganic sulphur, percent} \\ \text{by mass} \end{array} = \frac{A \times 0.1373}{B} \times 100$$

where

A = mass in g of barium sulphate, and

B = mass in g of specimen used.

A-6. ORGANIC SULPHUR

A-6.1 Organic sulphur is calculated by subtracting inorganic sulphur as obtained in A-5 from total sulphur as obtained in A-3 or A-4 as applicable.

A-7. FREE SULPHUR

A-7.1 General — This method covers the determination of the amount of free sulphur in rubber products. It is applicable to NR, SBR, BR, NBR, CR and IR products.

A-7.2 Apparatus

A-7.2.1 Buchner Funnels, prepared by forming a thin asbestos pad, or a thicker layer of a diatomaceous earth filter-aid, over a single sheet of qualitative filter paper. Filters thus prepared can be used numerous times.

A-7.3 Reagents and Materials

A-7.3.1 Cadmium Acetate Solution (30 g/dm³)

A-7.3.2 Cadmium Acetate Wash Solution (1.2 g/dm³)

A-7.3.3 Formaldehyde Solution (40 percent)

A-7.3.4 Glacial Acetic Acid

A-7.3.5 Iodine, Standard Solution (0.05 N) — Add 6.35 g of iodine and 20 g of potassium iodide (KI) to a beaker and just cover with water. Let stand with occasional stirring until dissolved, adding a small additional amount of water if necessary. When dissolved, dilute to 1 dm³, filter through a filter crucible, and store the solution in a stoppered, brown glass bottle. Standardize, preferably on the day it is to be used, against standard arsenic trioxide.

A-7.3.6 Paraffin

A-7.3.7 Sodium Stearate Suspension in Water (1 g/dm³)

A-7.3.8 Sodium Sulphate Solution — (50 g Na₂SO₃/dm³.)

A-7.3.9 Starch Solution — (10 g/dm³).

A-7.3.10 Strontium Chloride Solution — (5 g SrCl₂/dm³.)

A-7.4 Procedure

A-7.4.1 Place 2 g of a sample thinly sheeted (0.5 to 0.75 mm) in a 400 cm³, thin-walled, chemically resistant glass flask. Add 100 cm³ of Na₂SO₃ solution, 5 cm³ of sodium stearate suspension in water, and approximately 1 g of paraffin. Cover the flask with a small watch glass and heat so as to boil gently for 4 h, or digest just below the boiling point for 16 h. Remove the flask and add 100 cm³ of SrCl₂ solution and 10 cm³ of cadmium acetate solution. Separate the rubber and precipitate by filtration, using a Buchner funnel with suction (see A-7.2.1). Wash with two 75 to 100 cm³ portions of cadmium acetate wash solution.

A-7.4.2 To the filtrate, add while stirring to 10 cm³ of formaldehyde solution, 10 cm³ of glacial acetic acid, and 5 cm³ of starch solution. Add enough crushed ice to bring the temperature of the solution below 15°C, and titrate with 0.05 N iodine solution to a blue end point.

A-7.5 Blank — The blank determination on the reagents should not usually exceed 0.2 to 0.3 cm³.

A-7.6 Calculation — Calculate the percentage of free sulphur as follows:

$$\text{Free sulphur, percent} = (A - B) N \times 0.032/C \times 100$$

where

A = volume of iodine solution required for titration of the sample, cm³;

B = volume of iodine solution required for titration of the blank, cm³;

N = normality of the iodide solution; and

C = grams of sample used.

A-8. FILLERS (REFEREE ASH METHOD)

A-8.1 General — This method is intended for settling disagreements on fillers content by the ashing method. It is also useful for ashing rubber products for determination of inorganic sulphur in the absence of antimony or for ash analysis.

A-8.1.1 The method is not accurate for rubbers containing halogens when zinc compounds or other metal compounds that form volatile halides are present and shall not be used if an analysis of these metals in the ash is required.

A-8.1.2 The method may be used for preparing samples for ash analysis in rubber products not containing halogens or antimony sulphide. However, if the sample contains carbonates that decompose at 550°C or clays or silicates that will lose water at this temperature, the ash content value will not be highly precise and will not represent the original amount the ash content of inorganic fillers present in the sample.

A-8.2 Apparatus

A-8.2.1 Crucibles — porcelain or silica, approximately 50 ml capacity.

A-8.2.2 Electric Muffle Furnace — with controls necessary to hold the temperature at $550^{\circ}\text{C} \pm 25^{\circ}\text{C}$.

A-8.2.3 Calibrated Thermocouple and Temperature Recording Device

A-8.3 Procedure

A-8.3.1 Weigh a 1 g specimen of the sample into an ignited, weighed crucible.

A-8.3.2 Adjust the temperature of the muffle furnace to $550^{\circ}\text{C} \pm 25^{\circ}\text{C}$, place the crucible in the furnace, and close the door completely. When more than one crucible is to be placed in the furnace, the crucible shall be placed on a tray and put into the furnace. The door shall be closed immediately and not opened for 1 h. After 1 h, open the furnace door 3 to 5 cm and continue heating for 30 minutes or until all carbonaceous material is burned off.

NOTE — If a referee ash determination is required, the temperature shall be determined by placing the calibrated thermocouple sensing element at approximately the geometric centre of the furnace cavity. The temperature shall be adjusted to be within the specified range. No more than two crucibles shall be placed in the oven and they shall be positioned directly below the thermocouple.

A-8.3.3 Remove the crucible from the furnace, cool in a desiccator, and weigh.

A-8.4 Calculation

$$\text{Ash, percent by mass} = \frac{A - B}{C} \times 100$$

where

A = mass in g of ash and crucible,

B = mass in g of crucible, and

C = mass in g of specimen.

A-9. CARBON BLACK (NITRIC ACID DIGESTION METHOD)

A-9.1 General

A-9.1.1 This method covers the determination of carbon black by a nitric acid digestion method. It is applicable only to the 'R' family of rubbers containing an unsaturated carbon chain.

A-9.1.2 Application to any other rubber type may be possible if the rubber is degraded to fragments soluble in water, acetone, or chloroform. The method shall be used on such other rubbers only if adequate testing of known compounds has demonstrated the usefulness of the method.

A-9.2 Reagents

B-9.2.1 *Acetone*

A-9.2.2 *Chloroform*

A-9.2.3 *Sodium Chromate Solution* — 100 g/dm³.

A-9.2.4 *Sodium Hydroxide Solution* — 175 g/dm³ and 300 g/dm³.

A-9.3 Procedure

A-9.3.1 Extract a 0.5 g specimen with acetone-chloroform mixture in accordance with A-1.

A-9.3.2 Transfer the specimen to a 250 cm³ beaker and heat on the steam bath until it no longer smells of chloroform. Add a few cm³ of nitric acid and allow to stand for about 10 minutes. Add 50 cm³ more of nitric acid, taking care to wash down the sides of the beaker, and heat on the steam bath for at least 1 h. At the end of this time there should be no more bubbles or foam on the surface. Pour the liquid, while hot, into a Gooch crucible, taking care to keep as much as possible of the insoluble material in the beaker. Filter slowly with gentle suction and wash well by decantation with hot nitric acid. (*Caution* — Empty the filter flask) (*see Note*). Wash with acetone and a mixture of equal parts of acetone and chloroform until the filtrate is colourless. Digest

the insoluble material which has been carefully retained in the beaker, for 10 minutes on the steam bath with 35 cm³ of sodium hydroxide solution (300 g/litre). This treatment with alkali may be omitted if silicates are absent. Dilute to 60 cm³ with hot water and heat on the steam bath. Filter the solution of alkali and wash well with hot sodium hydroxide solution (175 g/dm³).

NOTE — The filtration may be materially aided, particularly with some synthetic rubber products, by partial or complete neutralization of the nitric acid solution with ammonium hydroxide. Partial neutralization together with the addition of trivalent cations or anions may also aid agglomeration of the carbon black particles if they are too well dispersed to filter.

A-9.3.3 Next, wash the residue about four times with hot hydrochloric acid. Neutralize the last washing with ammonium hydroxide and test for the presence of lead with sodium chromate solution. If lead is present, continue to wash with hot hydrochloric acid and, finally, wash with warm hydrochloric acid (1 : 7). Remove the crucible from the funnel, taking care that the outside is perfectly clean, dry it in an air bath for 1½ h at 110°C, cool and weigh; call this mass *a*. Burn off the carbon at a dull red heat (550 to 600°C) and reweigh; call this mass *b*. The difference in mass represents approximately 105 percent of the carbon originally present in the form of carbon black.

A-9.4 Calculation

$$\text{Carbon black, percent by mass} = \frac{a - b}{1.05 \times c} \times 100$$

where *c* is mass in g of specimen used.

A-10. CARBON BLACK (PYROLYSIS METHOD)

A-10.1 General

A-10.1.1 This method covers the determination of carbon black by a pyrolysis method. It is applicable to all 'R' and 'M' family rubbers except those containing halogens or nitrogen in the rubber polymer or lead salts or phenolic resins as added components.

A-10.1.2 Application to other rubbers may be possible if no carbonaceous residue is formed under the pyrolysis conditions. The method shall be used on such other polymers only if adequate testing of known compounds has demonstrated the usefulness of the method.

A-10.1.3 In this method the sample is heated in a furnace at about 1 000°C in an atmosphere of nitrogen to distil all other materials except ash and free carbon. Ash is determined separately and from the difference of the two, amount of carbon black is obtained.

A-10.2 Apparatus

A-10.2.1 Electrical Combustion Furnace — Tubular in shape, about 325 mm in length and having a heating zone of about 275 mm, and capable of maintaining a temperature of $1\,000^{\circ}\text{C} \pm 25^{\circ}\text{C}$.

A-10.2.2 Combustion Tube — of transparent silica, about 725 mm in length, 26 and 28 mm in inner and outer diameter respectively.

A-10.2.3 Nitrogen Gas Cylinder

A-10.2.4 Gas Washing Bottles

A-10.2.5 Glazed Porcelain Boat

A-10.3 Procedure

A-10.3.1 Ignite the boat and weigh it after cooling. Accurately weigh, in the boat, about 1 g of the homogenized sample prepared as in 3.1.1 (see page 4). Heat the furnace with the combustion tube to approximately $1\,000^{\circ}\text{C}$. Draw out the combustion tube a little bit from the hot zone and clean it with nitrogen gas for 1 minute to drive away the air. Introduce the weighed boat with sample into the combustion tube, connect the gas flow and push the tube so that the boat is in the hot zone of the furnace. At first dense, black-brown smoke will come out of the combustion tube which will slowly turn to white smoke. When white smoke becomes thinner, draw out the combustion tube from the furnace. Continue the flow of nitrogen for 2-3 minutes more. Cut off the supply of nitrogen and take out the boat carefully avoiding spilling and keep it in a desiccator till it is cooled. Weigh the boat with the ignited sample.

NOTE — Nitrogen gas used shall be cleaned by passing it through washing bottles containing glycerine and alkaline pyrogallol (20-30 g of pyrogallol in 10 ml of about 30 percent potassium hydroxide) respectively to remove the moisture and traces of oxygen present in nitrogen.

A-10.4 Calculation

A-10.4.1 Calculate the carbon black content as follows:

$$\text{Total fillers, percent by mass} = (A) = \frac{M_3 - M_1}{M_2 - M_1} \times 100$$

where

M_3 = mass in g of the boat with the sample after ignition,

M_1 = mass in g of empty boat, and

M_2 = mass in g of the boat with sample.

A-10.4.1.2 Ash content, percent by mass (calculated as given in **A-8.3**) = B .

A-10.4.1.3 Carbon black, percent by mass = $A - B$.

A-11. NITROGEN (CALCULATED AS GLUE)

A-11.1 General — This method is intended for use in the determination of glue when it is used as a filler in rubber products other than NBR products. The method may also be used to determine nitrogen from NBR polymer present in a product. The calculations in this section are designed for determination of glue in NR products. In the absence of other nitrogenous material, the method may be applied to synthetic rubber products without correction for the nitrogen content of the rubber polymer. A slight error will result from nitrogen content of antioxidants and accelerators present after extraction. When glue is found to be present in a NR product and determined by this method by calculation from nitrogen content, a correction shall be made for the natural protein in NR as shown in the calculations. The correction shall be made by the method for approximation. The rubber as compounded is calculated on the basis of the total nitrogen as glue. The glue content is then corrected on the assumption that the rubber as compounded contains 0.4 percent protein nitrogen and this value is used to calculate a more exact figure for rubber hydrocarbon.

A-11.2 Reagent

A-11.2.1 *Copper Sulphate* — (CuSO_4)

A-11.2.2 *Paraffin*

A-11.2.3 *Sodium Hydroxide Solution* — 750 g/dm³.

A-11.2.4 *Sodium Hydroxide Standard Solution* — Prepare and standardize a 0.1 N sodium hydroxide solution.

A-11.2.5 *Sodium Sulphate* — (Na_2SO_4)

A-11.2.6 *Sulphuric Acid, Standard Solution* — Prepare and standardize a 0.1 N sulphuric acid solution.

A-11.2.7 *Zinc* — granulated.

A-11.3 Procedure — Extract a 2 g specimen with acetone for 8 hours. Remove the solvent from the specimen, and transfer the latter from the filter paper to a 750 cm³ Kjeldahl flask. Add 25 to 30 cm³ of sulphuric acid, 10 to 12 g of sodium sulphate and about 1 g of copper sulphate.

Heat gently until the first vigorous frothing ceases; then raise the heat gradually until the liquid boils. Continue the boiling until the solution becomes clear. Allow the flask to cool, dilute carefully with 150 cm³ of water, and again allow to cool. Add 100 cm³ of sodium hydroxide solution (750 gdm³), pouring it carefully down the side of the flask so that it does not mix immediately with the acid solution. Add about 1 g of granulated zinc to prevent bumping and a piece of paraffin of the size of a pea to diminish frothing. Connect the flask quickly with a condenser, the delivery tube of which dips into a 500 cm³ conical flask, containing 50 cm³ of 0.1 N sulphuric acid diluted to about 100 cm³. Carefully swirl the flask to mix the contents and start to heat gently; increase the flame as the danger of foaming over diminishes, and finally boil briskly until about one-half of the liquid has passed over into the receiver. Add methyl red solution and titrate the excess acid by means of 0.1 N sodium hydroxide solution.

A-11.4 Calculation

$$\text{Total nitrogen as glue, percent by mass} = \frac{(AN - BN') \times 0.014 \times 6.5}{C} \times 100$$

where

A = volume in cm³ of sulphuric acid used,

N = normality of sulphuric acid,

B = volume in cm³ of sodium hydroxide used,

N' = normality of sodium hydroxide, and

C = mass in g of specimen used.

A-12. RUBBER POLYMER CONTENT

A-12.1 Calculation

Rubber polymer content, percent = $A (100 - B)$

where

A = factor as listed below:

<i>Rubber</i>	<i>Factor A</i>
NR	94/97
IR	1.00

SBR *	1.00
BR	1.00
IIR	1.00

B = Sum of percentage, of total extract, alcoholic potash extract, organic sulphur, ash, carbon black and glue determined in accordance with **A-1, A-2, A-6, A-7, A-8** and **A-9** respectively.

*Containing 23.5 percent bound styrene and not oil extended.

(Continued from page 2)

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